

1. What is Advanced Java?

Advanced Java is used to develop:

- Web applications
- Enterprise applications
- Distributed systems

It includes:

- JDBC
- Servlets
- JSP
- Frameworks

2. JDBC (Java Database Connectivity)

JDBC is used to connect Java with database.

Steps:

```
import java.sql.*;
```

```
Connection con = DriverManager.getConnection("url","user","pass");
```

```
Statement stmt = con.createStatement();
```

```
ResultSet rs = stmt.executeQuery("SELECT * FROM users");
```

Explanation:

- Connection → connects database
- Statement → executes query
- ResultSet → stores result

Use:

Database operations

3. Types of JDBC Drivers

- Type 1 → JDBC-ODBC
- Type 2 → Native API
- Type 3 → Network Protocol
- Type 4 → Thin Driver

Use:

Different ways to connect DB

4. Servlet

Servlet is a Java class used to handle HTTP requests.

```
import javax.servlet.*;
import javax.servlet.http.*;

public class MyServlet extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse
res) {
        res.getWriter().println("Hello");
    }
}
```

Use:

Handle request-response

5. Servlet Life Cycle

- `init()` → initialization
- `service()` → request handling
- `destroy()` → cleanup

Use:

Manage servlet execution

6. HTTP Methods

- `GET` → retrieve data
- `POST` → send data

Use:

Communication between client & server

7. JSP (Java Server Pages)

JSP is used to create dynamic web pages.

```
<%  
out.println("Hello JSP");  
%>
```

Use:

Embed Java in HTML

8. JSP Elements

- Scriptlet `<% %>`
- Expression `<%= %>`
- Declaration `<%! %>`

Use:

Write Java code in JSP

9. JSP Directives

```
<%@ page language="java" %>
```

Use:

Provide instructions

10. Session Management

```
HttpSession session = request.getSession();  
session.setAttribute("user", "John");
```

Use:

Store user data

11. Cookies

```
Cookie c = new Cookie("user", "John");  
response.addCookie(c);
```

Use:

Store data in browser

12. Filters

Filters process requests before reaching servlet.

```
public void doFilter(...) {  
}
```

Use:

Authentication, logging

13. Listeners

Listeners respond to events.

```
public class MyListener implements ServletContextListener {}
```

Use:

Event handling

14. MVC Architecture

- Model → Data
- View → UI
- Controller → Logic

Use:

Organize application

15. JavaBeans

```
public class User {  
    private String name;  
    public String getName() { return name; }  
}
```

Use:

Reusable components

16. JDBC Prepared Statement

```
PreparedStatement ps = con.prepareStatement("INSERT INTO  
users VALUES(?");
```

Use:

Prevent SQL injection

17. Transactions

```
con.setAutoCommit(false);
```

Use:

Control database operations

18. Connection Pooling

Use:

- Reuse database connections
- Improve performance

19. JSP Implicit Objects

- request
- response
- session
- application

Use:

Access data easily

20. JSTL (Java Standard Tag Library)

```
<c:out value="Hello"/>
```

Use:

Simplify JSP

21. Expression Language (EL)

```
${user.name}
```

Use:

Access data

22. Web.xml Configuration

```
<servlet>
  <servlet-name>MyServlet</servlet-name>
</servlet>
```

Use:

Configure web app

23. Deployment

Use:

- Compile
 - Package (WAR file)
 - Deploy on server (Tomcat)
-

24. Apache Tomcat Server

Use:

- Run Java web apps
 - Supports servlets & JSP
-

25. Security Basics

Use:

- Authentication
- Authorization
- HTTPS

26. Exception Handling

```
try {  
    int x = 10/0;  
} catch(Exception e) {  
    System.out.println("Error");  
}
```

Use:

Handle runtime errors

27. Logging

Use:

- Track errors
 - Debug application
-

28. Framework Introduction

Popular frameworks:

- Spring
- Hibernate

Use:

Simplify development